

Refine Judgement About Reasonableness and Accuracy

Step 12

Upper Elementary



Today I am learning to judge the reasonableness and accuracy of measurements and explain my thinking.

1. Does the Answer Make Sense?

For each statement, decide if the answer makes sense. Circle your choice.

- a)  A pencil is 15 cm long.
☐ Makes Sense ☐ Does Not Make Sense
- b)  A dog weighs 250 kg.
☐ Makes Sense ☐ Does Not Make Sense
- c)  A classroom is 8 m long.
☐ Makes Sense ☐ Does Not Make Sense
- d)  A drink bottle holds 600 L.
☐ Makes Sense ☐ Does Not Make Sense

2. Spot the Measurement Mistake

Read each situation and explain what is wrong.

- a) Sam measured a pencil as 2 m long.
 What is wrong?


- b) Ella said a swimming pool holds 50 mL.
 What is wrong?


- c) Ben measured a watermelon as 300 kg.
 What is wrong?


3. Estimate and Check

Estimate, then measure and record. Find the difference.

- a) Length of your pencil 
 Estimate: _____ cm
 Actual: _____ cm
 Difference: _____ cm
- b) Width of your desk 
 Estimate: _____ cm
 Actual: _____ cm
 Difference: _____ cm

4. Which Is More Accurate?

Circle the more accurate measurement.

- a) 5 m or 5.23 m
- b) 2 L or 2.15 L
- c) 20 cm or 20.4 cm
- d) 4 kg or 4.35 kg

5. Choose the Best Measurement

Choose the most appropriate unit.

- a) Length of a classroom 
☐ cm ☐ m ☐ km
 Explain your choice: _____
- b) Mass of a dog 
☐ g ☐ kg ☐ t
 Explain your choice: _____
- c) Capacity of a bucket 
☐ mL ☐ L ☐ kL
 Explain your choice: _____

6. Measurement Investigation

A student measured the length of a book three times.



The measurements were:

21 cm 22 cm 21.5 cm

- a) Which measurement is likely to be most accurate?

- b) Why do you think that?

- c) What could cause the different measurements?

7. Real-Life Problem Solving

A recipe needs 2.5 L of water. You only have a 1 L jug.



- a) How many full jugs of water will you need?

- b) Will one more full jug be too much?

- c) How could you measure exactly 2.5 L?

Helpful thinking: 

- 2.5 L is more than 2 L.
- It is less than 3 L.
- You may need to measure carefully.

8. Reflect and Explain

- a) Why is accuracy important in measurement?


- b) Why should we check whether answers are reasonable?


- c) How has your measurement thinking improved?




I can...

- ☐ judge if measurements are reasonable
- ☐ identify and explain mistakes
- ☐ choose appropriate units
- ☐ estimate, measure and compare
- ☐ explain my measurement reasoning

Measurement Words



Great job!

You are becoming a careful and confident measurement thinker!